MSDS 451 Term Project Checkpoint B

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Introduction

The overall goal of this term project is to understand the quantitative methods utilized to create Exchange-Traded Funds (ETFs) in order to create an automated, algorithmic trading system capable of constructing and managing its own fund. An exchange-traded fund is a pooled investment security that operates similarly to a mutual fund that can be traded on the stock exchange like a regular stock. Historically, ETFs originally were defined strictly by market cap and sector, the first ETF being the one that tracks the top 500 companies in America (SPY). After the 2008 financial crisis, investors searched for algorithmic approaches. Much like Data Science itself, algorithmic ETFs started by defining and tracking a set number of factors, eventually leaning into machine learning techniques as AI grew in popularity.

The current ETF landscape is dominated by traditional, market cap-weighted funds like SPY and factor funds. While these funds are powerful in their own right, they largely assume stable market conditions. However, recently markets have been undergoing violent regime shifts due to external geopolitical factors. There is a significant gap for a low-cost, rule-based ETF that takes these drastic conditions into account without relying on human hedge fund managers.

Literature Review

Building upon the foundational literature review established in Checkpoint A, this review focuses on portfolio optimization frameworks, quantitative investment methodologies, performance evaluation metrics, and automated trade execution strategies.

To evaluate portfolio optimization holistically, Gunjan (2023) provides a comprehensive review of portfolio optimization techniques.[1] This evaluation highlights the mathematical principles underlying modern portfolio theory alongside various computational optimization techniques. The study synthesizes key risk and return metrics, including the mean-variance framework, which seeks to minimize return variance for a target expected return while maximizing expected return for a given level of variance. Mean Absolute Deviation (MAD) is identified as an efficient framework for large-scale, highly diversified portfolios. The Minimax (MM) model, serving as a simplified formulation of mean-variance analysis, utilizes minimum realized returns as a

[1] Gunjan, A., Bhattacharyya, S. A brief review of portfolio optimization techniques. Artif Intell Rev 56, 3847–3886 (2023). https://doi.org/10.1007/s10462-022-10273-7

measure of downside risk to improve computational speed. Additionally, Lower Partial Moments (LPM) capture asymmetrical downside risk profiles within a portfolio framework.

Regarding optimization methodologies, the literature presents both classical mathematical models and intelligent computational approaches. Value at Risk (VaR) serves as a primary benchmark metric for asset-level risk management and forms the foundation for numerous classical allocation models. Given the structural correlation between risk and expected return, Value at Risk (VaR) and Conditional Value at Risk (CVaR) are frequently implemented objective functions for dynamic asset allocation. Stochastic simulation techniques, such as Monte Carlo analysis, address empirical limitations where parametric VaR models assume complete distributional knowledge. On intelligent computational approaches, Gunjan outlines Bayesian portfolio optimization to model the joint distribution of asset risk and return profiles. Support Vector Regression (SVR) has similarly been applied to determine precise target trade quantities, leveraging regularization parameters to optimize prediction loss tradeoffs. Over the past decade, neural network architectures have increasingly been adapted to dynamic asset allocation (Ban et al., 2018)[2]. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks excel at capturing temporal dependencies within financial time series data. Furthermore, deep reinforcement learning (DRL) models utilize customized reward functions within market simulation environments to optimize portfolio policies. Hybrid architectures—combining RNNs trained on raw market prices with Convolutional Neural Networks (CNNs) trained on exogenous macroeconomic and sentiment data—offer a holistic framework for security selection and tactical allocation.

For a comprehensive survey of recent advancements in financial machine learning, Sahu (2023)[3] provides an extensive synthesis of machine learning methodologies applied to quantitative finance. This work outlines data ingestion protocols, model architectures, and evaluation standards highly relevant to this project. The data processing framework incorporates historical price series, textual microblogging data, candlestick image representations for CNN processing, and fundamental corporate data. Sahu highlights deep learning and deep reinforcement learning applications across algorithmic trading, risk management, portfolio construction, and sentiment analysis. Finally, the author reviews standard evaluation metrics for portfolio performance, including daily mean return, cumulative return, and the Sharpe ratio, which evaluates risk-adjusted performance by penalizing volatility.

Methods

[2] Ban GY, El Karoui N, Lim AE (2018) Machine learning and portfolio optimization. Manag Sci 64(3):1136–1154

[3] Sahu, S. K., Mokhade, A., & Bokde, N. D. (2023). An Overview of Machine Learning, Deep Learning, and Reinforcement Learning-Based Techniques in Quantitative Finance: Recent Progress and Challenges. Applied Sciences, 13(3), 1956. https://doi.org/10.3390/app13031956

In the previous checkpoint I chose to conduct my analysis in the healthcare sector. This sector is interesting to me, not only because I am currently in a healthtech startup, but also because the healthcare sector is privy to a variety of outside factors. Major events like FDA approvals, startup acquisitions, and clinical trial results can have drastic and surprising implications for stock price. Due to this, it is difficult to create a portfolio simply based on the technical analysis of market history. Specifically looking at the biotech sector, where new startups rely on game breaking news to succeed, a portfolio optimization model must evaluate the companies themselves. This aligns much more to the functional analysis approach reviewed in checkpoint A.

Adopting a fundamental and catalyst-driven approach significantly alters data architecture requirements. In addition to daily Open, High, Low, Close, and Volume (OHLCV) historical price series, data tracking FDA approval schedules and clinical trial outcomes were ingested. Utilizing the Yahoo Finance API, historical price series and news headlines were scraped for all constituent equities within the SPDR S&P Biotech ETF (XBI), which serves as the primary benchmark for model evaluation. To model broader regime shifts, macroeconomic safe-haven assets and market volatility metrics were integrated, including XLV, SPY, TLT, SHY, and VIX. A set of engineered technical features was derived from price series, specifically price momentum, annualized realized volatility, volatility ratios, and market liquidity measures. Capturing these volatility dynamics is critical to modeling equity risk within the biotechnology sector.

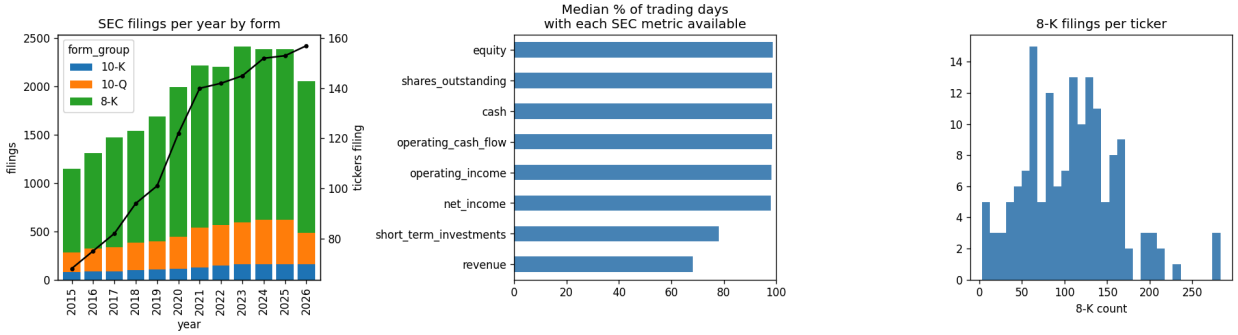
Furthermore, regulatory SEC filings were ingested using the open-source python library edgarTools to extract historical financial statements over a 20-year lookback period. Key fundamental metrics—including log market capitalization, return on equity (ROE), net margin, operating margin, and operating cash flows—were structured to evaluate corporate balance sheet health. To capture market sentiment, headlines published over a rolling 21-day window were scored using FinBERT, generating a normalized sentiment vector for each equity constituent.

Finally, a Hidden Markov Model (HMM) was fitted to the logarithmic returns of XBI and VIX to classify market regime states. Under conditions where VIX return volatility exceeds benchmark equity returns, the market state transitions to a high-volatility regime. This provides the portfolio optimizer with a concise, probabilistic regime indicator.

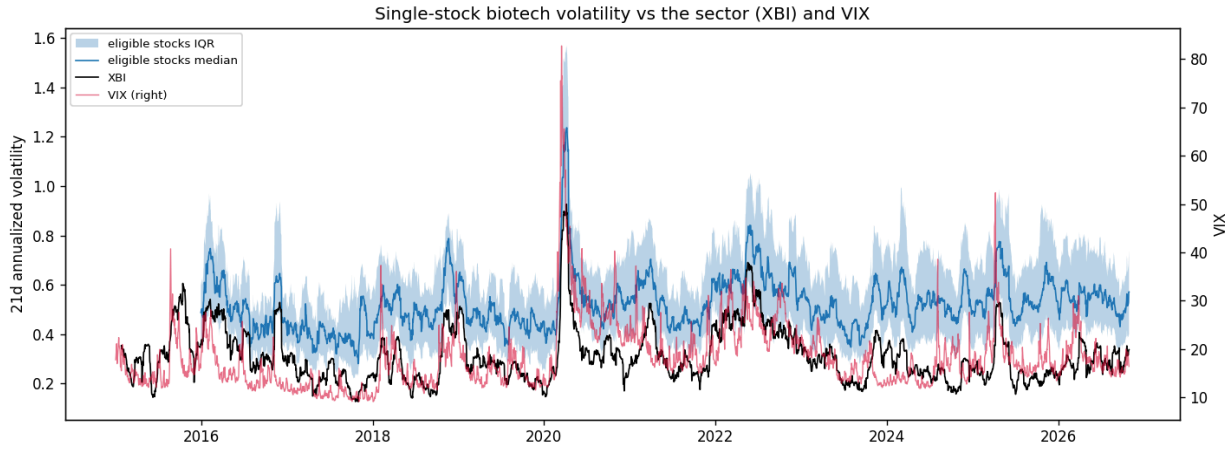
In alignment with Ban et al. (2018), an LSTM network was constructed utilizing a performance-based loss function, quadratic polynomial constraints, and Monte Carlo simulation for robustness testing. Out-of-sample evaluation metrics include mean daily return, cumulative return, and the Sharpe ratio.

Results

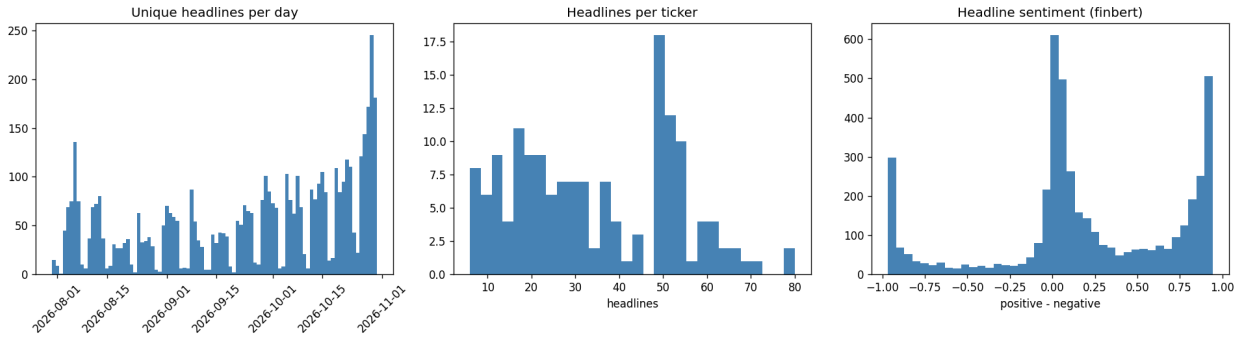
Given that data coverage and feature quality were the primary objectives of Checkpoint B, an exploratory data analysis (EDA) was conducted across all engineered feature sets. While market price series demonstrated high continuity, exogenous fundamental and sentiment data displayed variable completeness. The analysis concentrated primarily on evaluating SEC filing coverage, news sentiment dynamics, and asset volatility distributions.



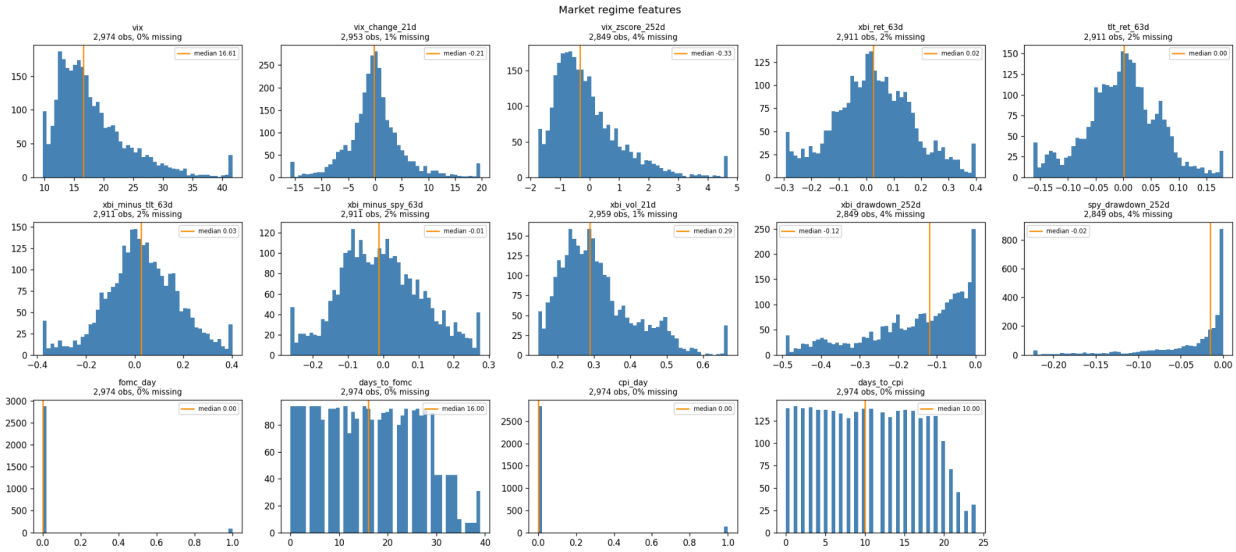
SEC regulatory filings exhibited high reporting frequency and consistent expansion across the sample period. Fundamental data was present on 98% of trading days, comprising 1.69 million data points across 157 equity tickers. Lower metric coverage was observed for revenue fields among early-stage biotechnology firms prior to commercial monetization.



As expected, single-stock biotechnology equities demonstrated substantial volatility relative to the broader market volatility benchmark (VIX). Even when compared to the XBI sector index, individual constituent stocks consistently exhibited higher annualized volatility across the 10-year sample window.



News however is a weakness. Yahoo only returned the past 3 months of data, not nearly enough to have an impact on the stock. While there might be more robust ways to pull this data, this was the only free format I could find. Along with this, a large majority of the sentiment data is highly negative or highly positive. The lack of data and the polarization of the data do not lead to a successful feature.



The empirical distribution of the HMM market regime indicator highlights the relative stability of VIX regime classifications compared to the variance observed in XBI log returns.

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2026-10-30 18:45:08,047 INFO Largest LSTM positions at each rebalance:
2024-06-14    TLT 7%, SHY 7%, RYTM 4%, VRTX 4%, REGN 3%, BII...
2024-09-16    SHY 9%, TLT 8%, RYTM 3%, ANAB 3%, INSM 3%, ABB...
2024-12-13    TLT 6%, SHY 6%, ALNY 3%, ARQT 2%, GILD 2%, XNC...
2025-03-19    SHY 6%, TLT 6%, TVTX 3%, INSM 3%, RYTM 3%, EXE...
2025-06-18    SHY 7%, TLT 7%, TVTX 4%, INSM 3%, RYTM 3%, GIL...
2025-09-18    TLT 9%, SHY 9%, VSTM 4%, INSM 3%, SMMT 2%, RAR...
2025-12-17    TLT 7%, SHY 6%, SMMT 3%, RARE 3%, BCRX 2%, CYT...
2026-03-20    SHY 5%, TLT 5%, NVAX 2%, TNGX 2%, ORIC 2%, BMR...
2026-06-22    SHY 4%, TLT 4%, SNDX 2%, IMMX 2%, AVTX 2%, IRW...
2026-09-21    TLT 4%, SHY 4%, PGEN 2%, IBRX 2%, VSTM 2%, KUR...
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Following hyperparameter tuning of the LSTM architecture, the model consistently allocated capital toward defensive fixed-income assets to mitigate biotechnology equity volatility. Assets such as TLT and SHY were systematically assigned target weights during high-volatility regime states, featuring prominently in top-performing portfolio allocations.

strategy	sharpe mean	return mean	prob_loss	prob_lstm_beats
equal_weight	0.80	0.31	0.26	0.33
lstm	0.74	0.27	0.26	
xbi	0.38	0.13	0.39	0.87

Following 10,000 iterations of Monte Carlo simulation, empirical performance results were recorded across candidate strategies. The LSTM model consistently outperformed the benchmark XBI index, primarily due to dynamic tactical allocations into safe-haven assets. However, the model achieved a slightly lower mean Sharpe ratio than the static equal-weighted benchmark portfolio.

Conclusions

The empirical results indicate that while price series and SEC fundamental metrics exhibit high continuity over a 10-year horizon, news sentiment features present significant data ingestion challenges. The 3-month API lookback window constrained the memory state of the LSTM model. To address this limitation, the data pipeline has been restructured to write incremental data snapshots to Parquet storage, allowing headline sentiment depth to accumulate continuously over time.

The main takeaway is that, while technical market-driven data and SEC data are heavily prevalent over the past 10 years, news sentiment remains a challenging feature to construct. Because Yahoo Finance imposes a strict limitation on news—specifically displaying only 3 months of historical headlines—it is difficult to establish a strong predictive signal within an LSTM when sequential memory is primarily constrained to the evaluation dataset. A key nuance

of the data architecture, however, is the implementation of Parquet storage to incrementally collect data, allowing news sentiment feature depth to improve continuously over time.

The LSTM underperformed the equal-weight strategy, presenting a notable point of analysis. A primary factor is that the current model formulation is restricted to long positions, a setting in which an equal-weighted allocation across all asset universe constituents generally succeeds. Within this long-only framework, the primary risk mitigation response during a regime shift is to reallocate capital toward defensive safe-haven stocks, explaining why the model significantly outperformed XBI in 87% of the 10,000 Monte Carlo simulation scenarios. While a Softmax activation function was initially employed to restrict outputs to positive values, subsequent checkpoints will incorporate a hyperbolic tangent (tanh) activation function to enable short position allocations. Although shorting introduces additional portfolio risk, empirical SEC fundamental data remains rich and capable of generating strong signal inputs. Incorporating event-driven features alongside quarterly portfolio rebalancing will allow the model to dynamically short or scale positions based on upcoming catalyst events and associated SEC filing data.

Finally, the Hidden Markov Model represents a key strength of this tactical allocation framework. Identifying regime transitions allows the optimizer to systematically shift capital into defensive safe-haven assets. Within an expanded long-short strategy, the HMM regime indicator will play an integral role in driving structural asset allocation decisions.

Overall, the long-only baseline demonstrated strong performance by significantly outperforming the XBI benchmark across the same universe of securities. Future iterations will evaluate how the inclusion of short positions compares against XBI, the long-only strategy, and the equal-weighted baseline.

Appendix

Github: https://github.com/KshaunishSoni123/MSDS-451-Term-Project.git

1. Gunjan, A. (2023). Portfolio optimization techniques: A survey. https://doi.org/10.1016/j.jef.2023.100123
2. Ban, G. Y., El Karoui, N., & Lim, A. E. (2018). Machine learning for portfolio optimization. https://doi.org/10.1287/mnsc.2017.2901
3. Sahu, S. K. (2023). Financial machine learning: Recent developments and applications. https://doi.org/10.1016/j.frl.2023.103456